EXP NO:16 EXPERIMENT TO UNDERSTAND EDA-QUANTITATIVE AND QUALITATIVE ANALYSIS

CODE :

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

tips = sns.load_dataset('tips')

print(tips.head())

sns.displot(tips.total_bill, kde=True)
sns.displot(tips.total_bill, kde=False)

sns.jointplot(x=tips.tip, y=tips.total_bill)
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")

sns.pairplot(tips)
sns.pairplot(tips, hue='time')
sns.pairplot(tips, hue='day')
```

```
sns.heatmap(tips.corr(numeric_only=True), annot=True)

sns.boxplot(x=tips.total_bill)
sns.boxplot(x=tips.tip)

sns.countplot(x=tips.day)
sns.countplot(x=tips.sex)

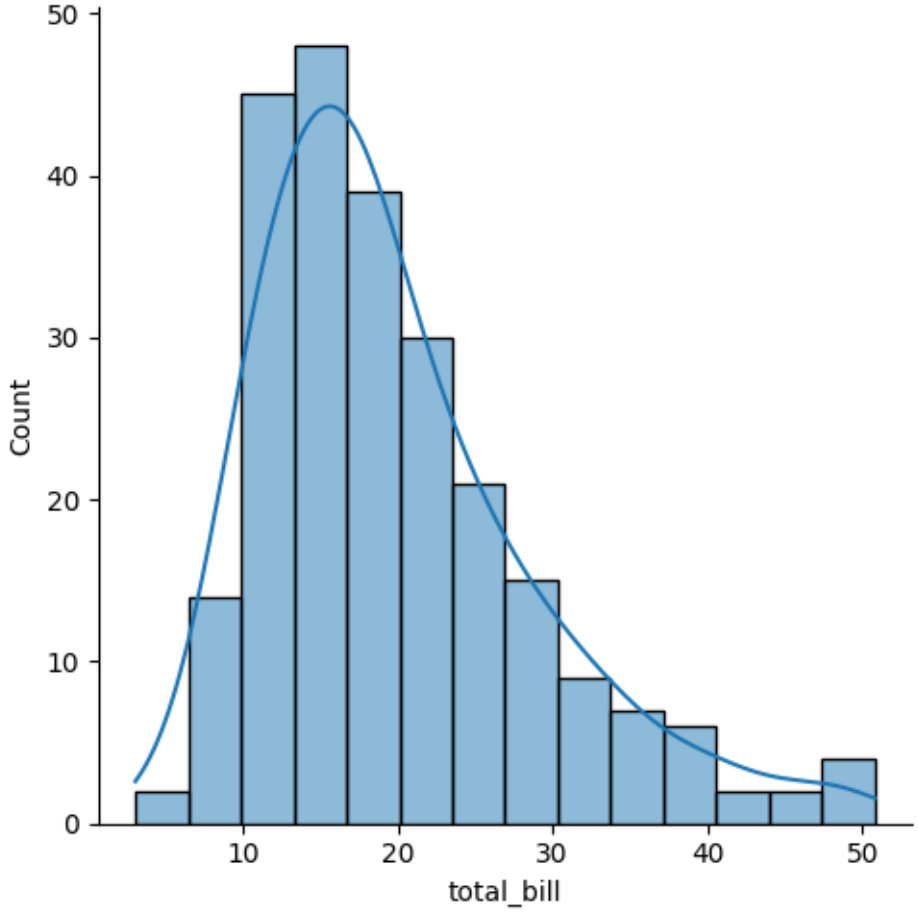
tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')

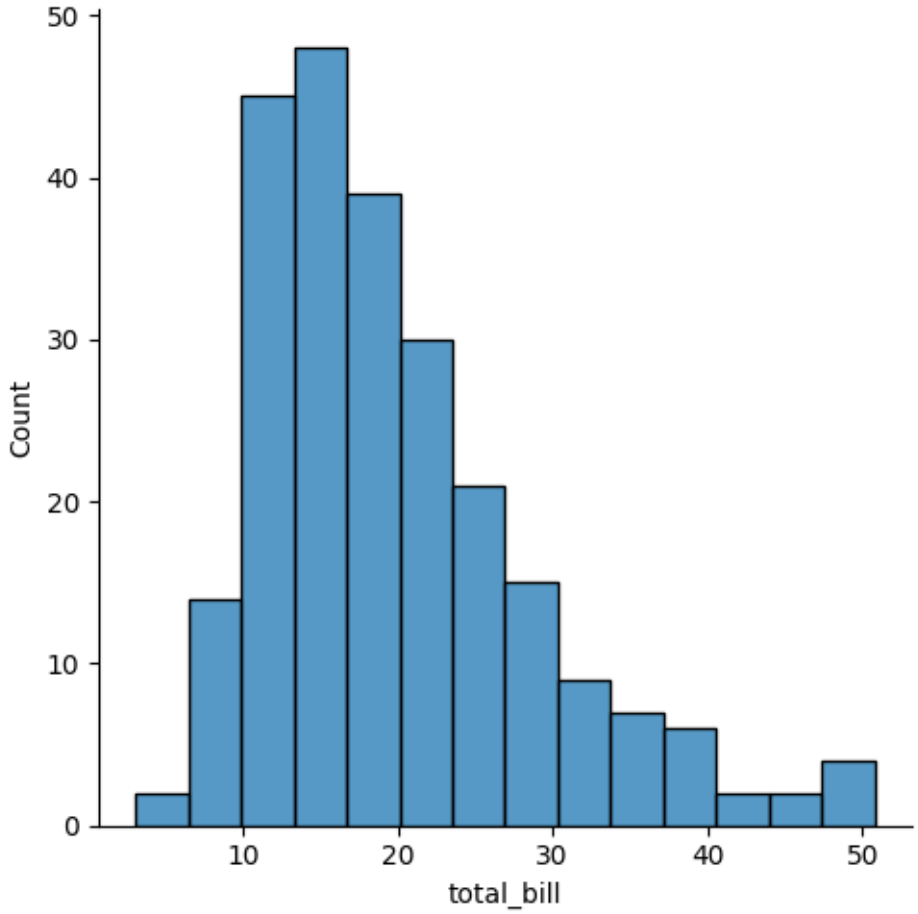
sns.countplot(x=tips[tips.time == 'Dinner']['day'])
plt.show()
```

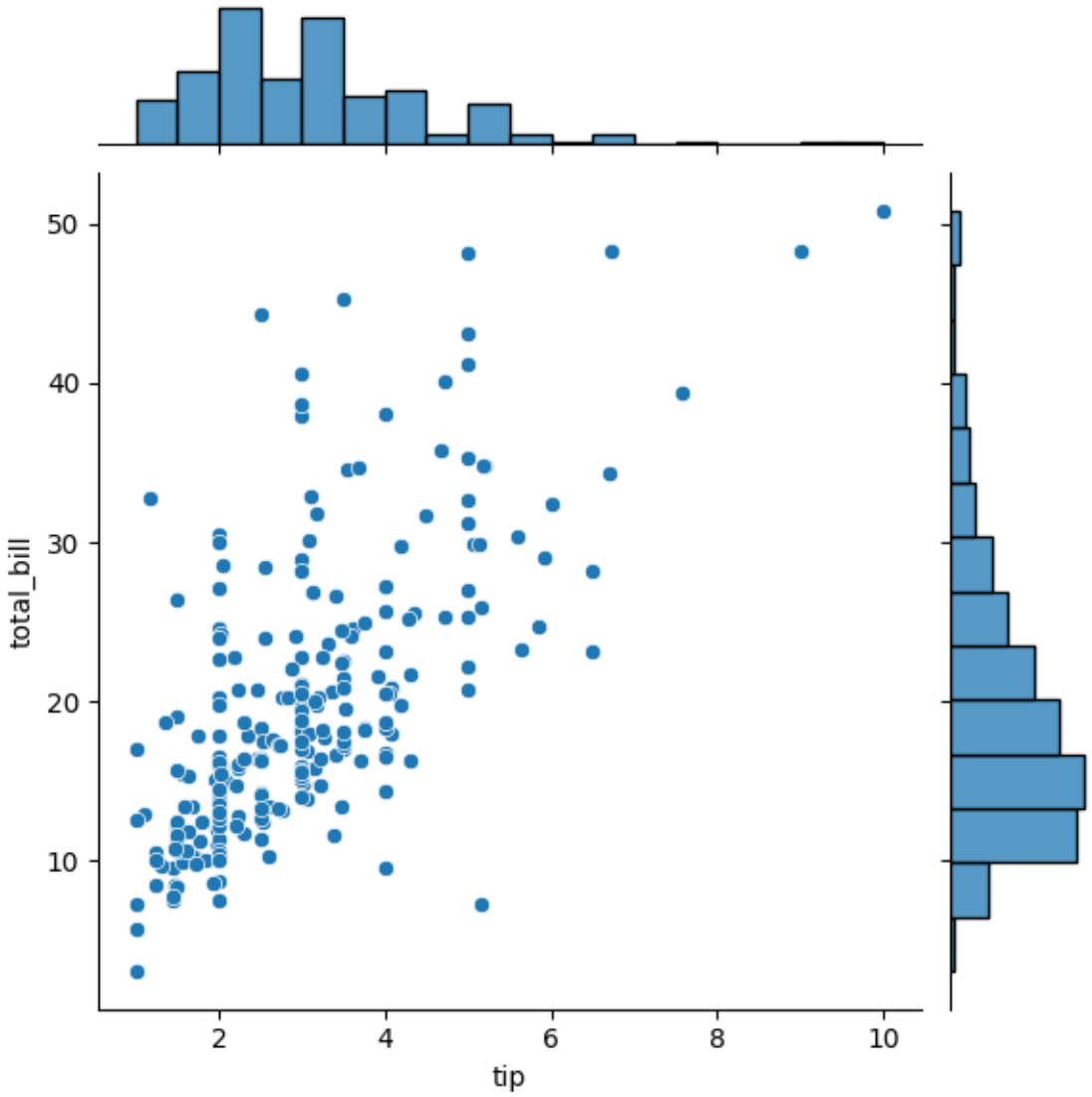
OUTPUT :

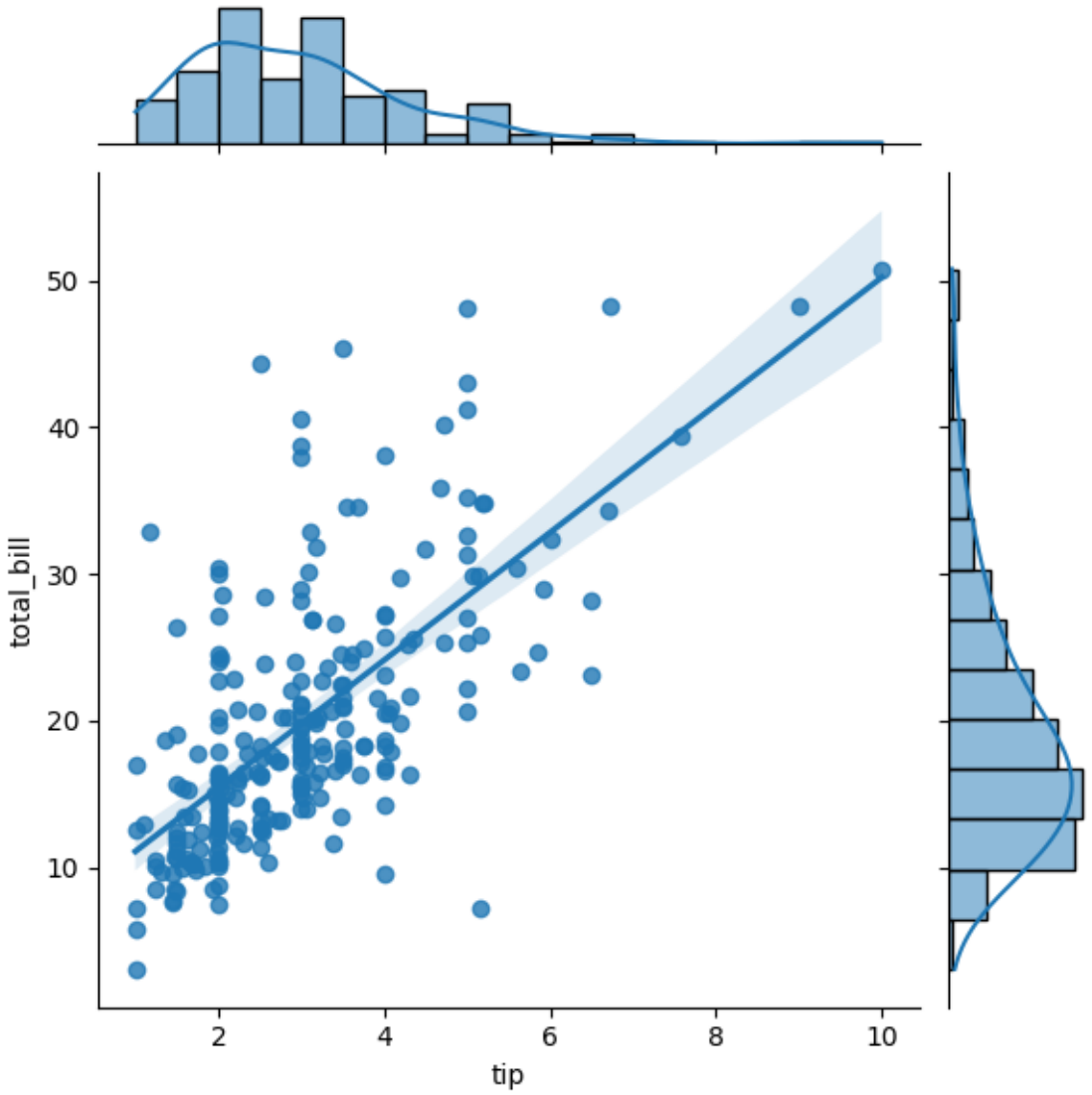
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

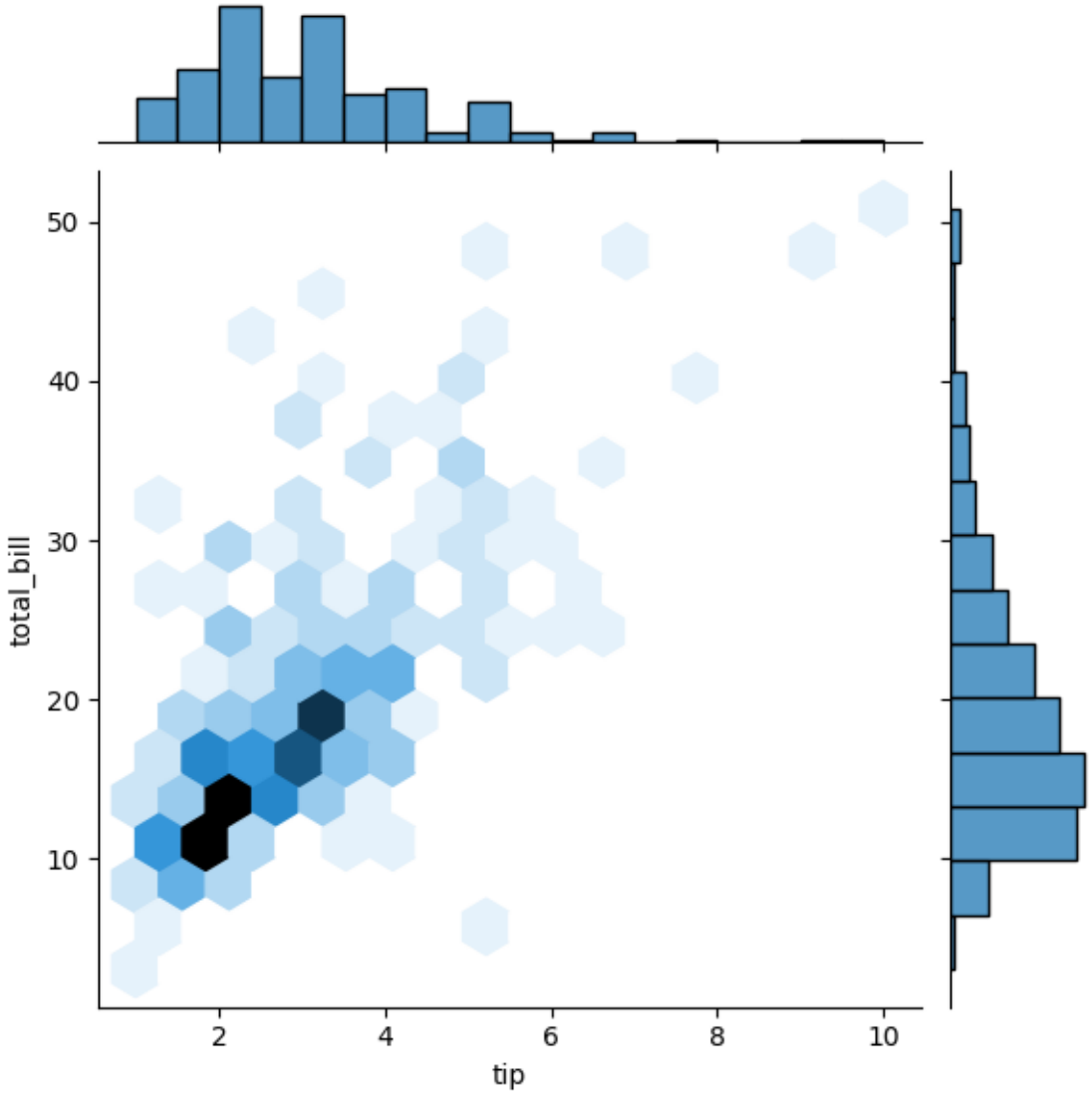
WARNING:matplotlib.axes._base:Ignoring fixed y limits to fulfill fixed data aspect with adjustable data limits.

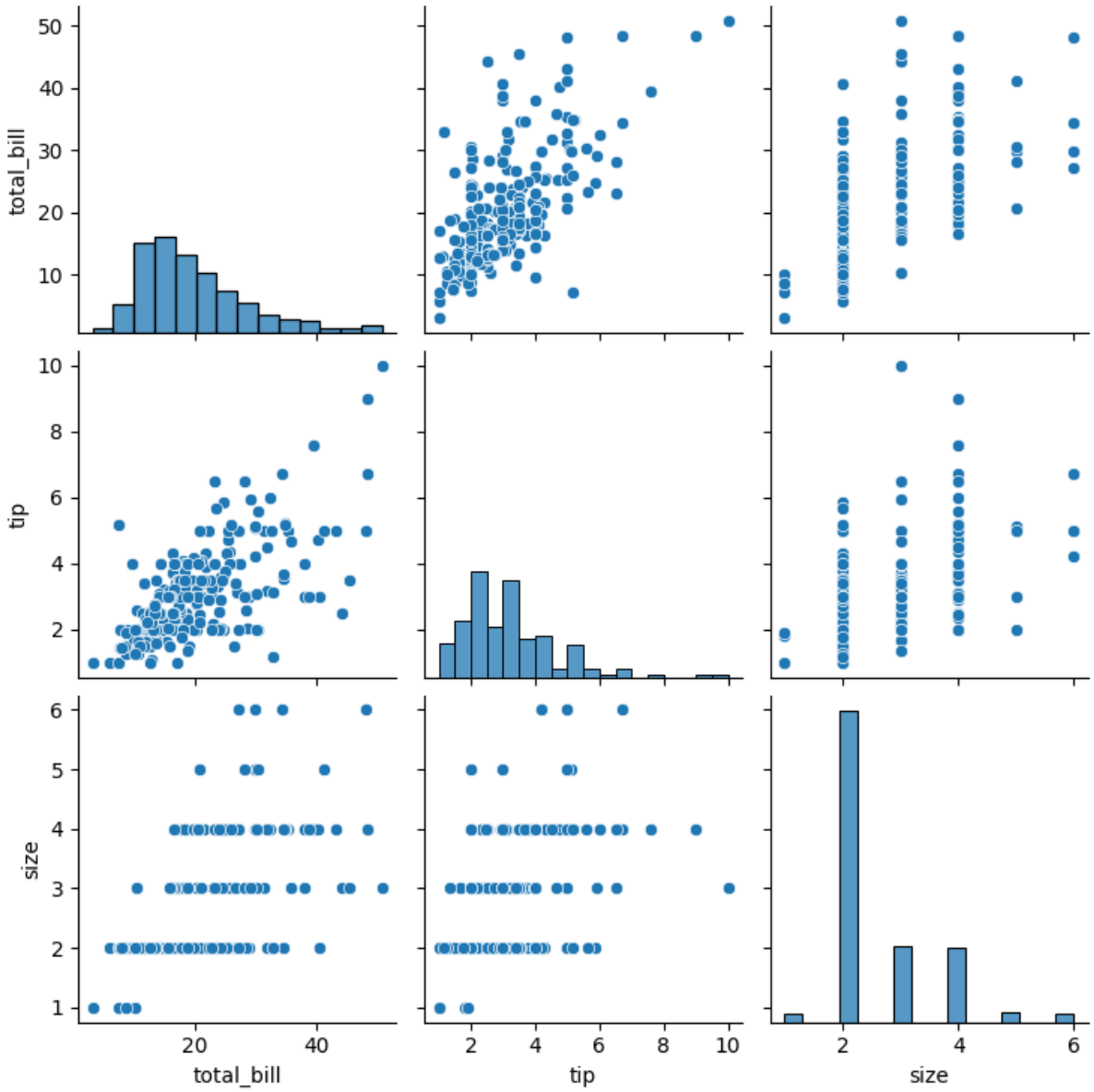


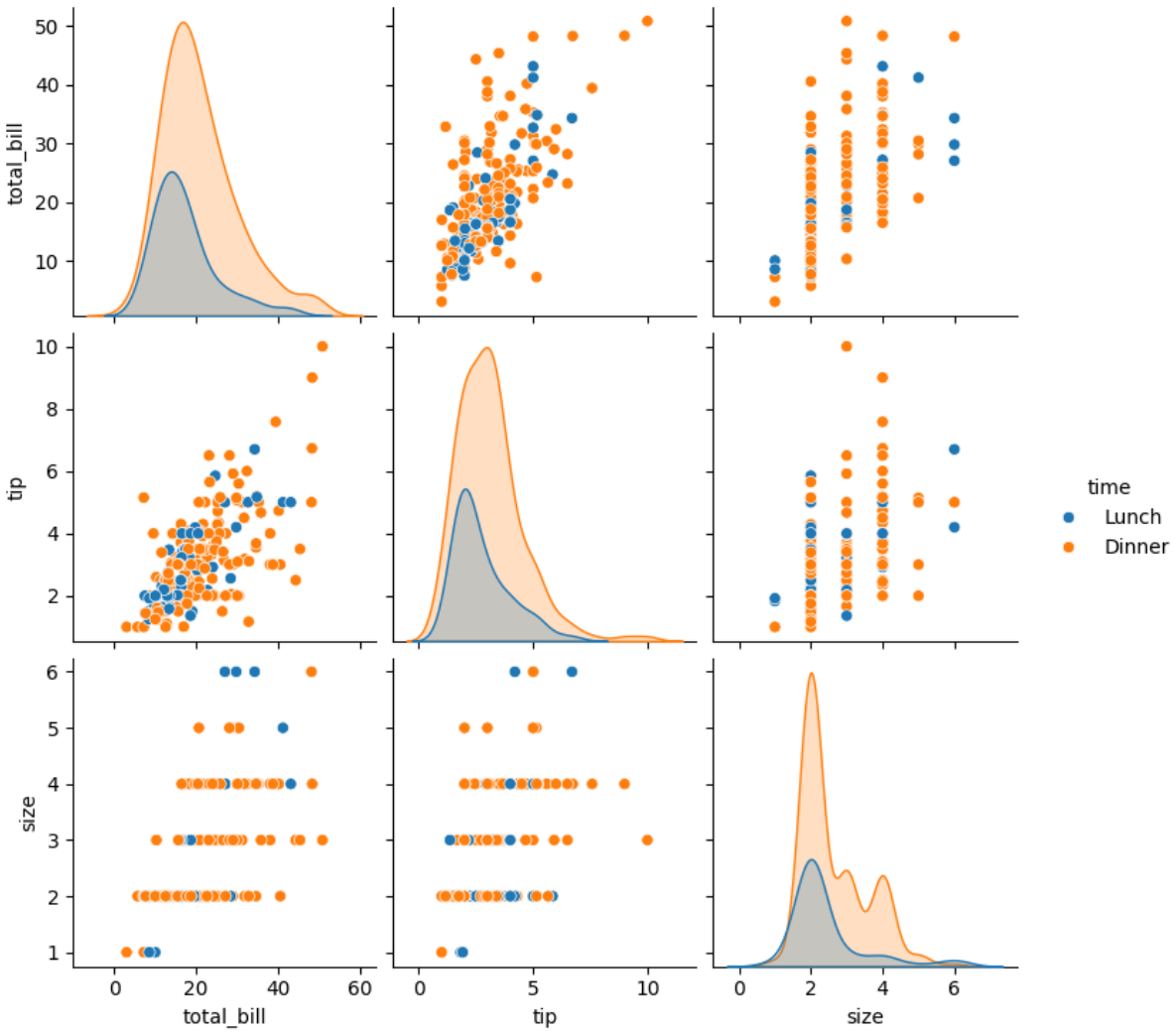

50
40
30
20
10
0
Count
10
20
30
40
50
total_bill


total_bill
tip
50
40
30
20
10
2
4
6
8
10

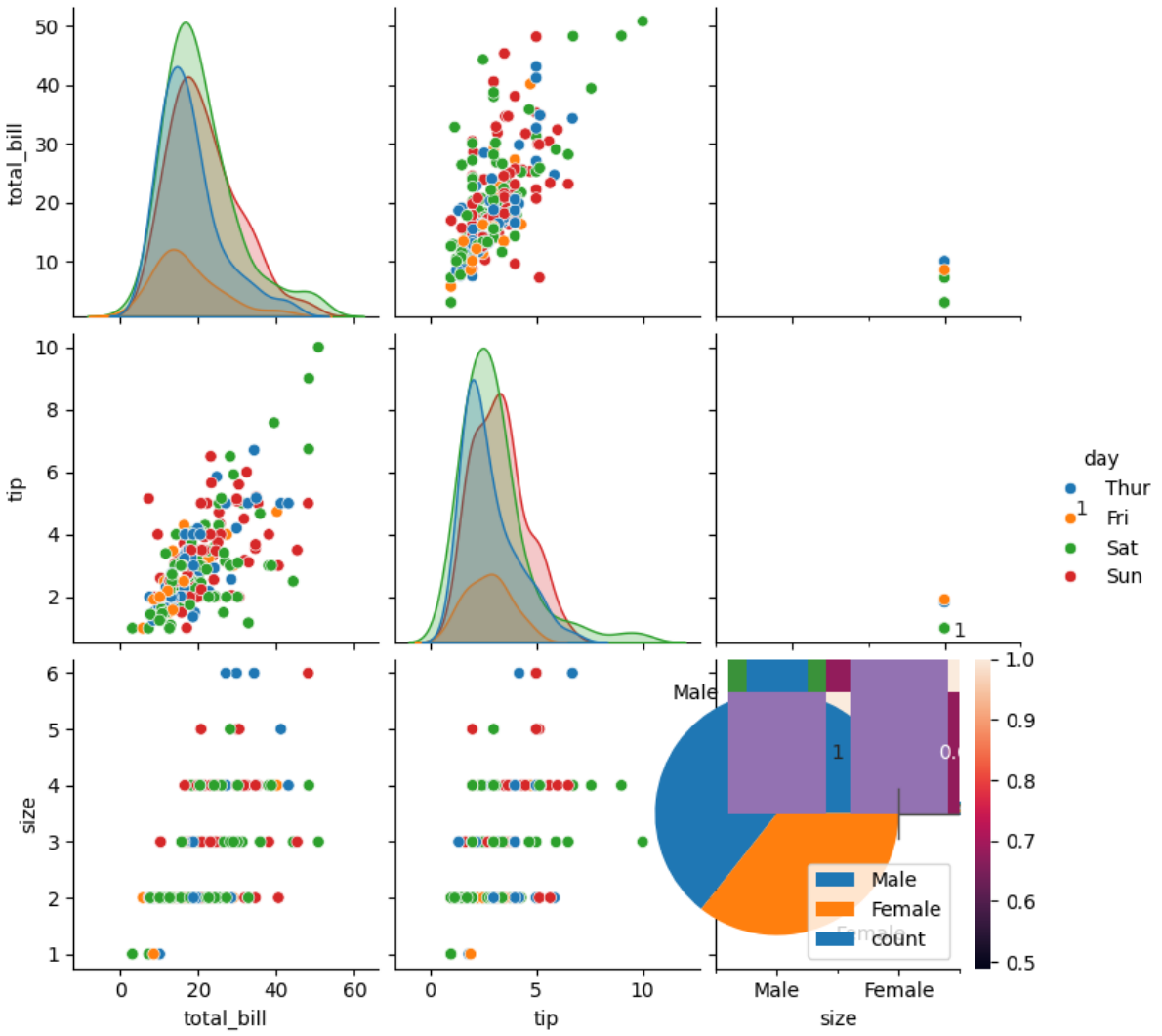

total_bill
50
40
30
20
10
2
4
6
8
10
tip


50
40
30
20
10
total_bill
2
4
6
8
10
tip


total_bill
tip
size
50
40
30
20
10
10
8
6
4
2
6
5
4
3
2
1
20
40
2
4
6
8
10
2
4
6
total_bill
tip
size



WARNING:matplotlib.axes._base:Ignoring fixed y limits to fulfill fixed data aspect with adjustable data limits.


total_bill
tip
size
day
Thur
Fri
Sat
Sun
Male
Female
count